

DISCRETE NEURAL REPRESENTATION LEARNING FOR PIANO AUDIO

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ABSTRACT

Discrete audio tokenizers for speech have been extensively studied, showing that certain tokens correspond to phonetic features. However, less is known about what features the learned tokens from discrete audio tokenizers from music encode. Previous work has focused on analyzing downstream classification tasks rather than directly analyzing the tokens themselves. To address this gap, we train a VQ-VAE with residual vector quantization (RVQ) on the MAESTRO dataset of piano audio and use the aligned MIDI annotations to evaluate what features the learned tokens encode by analyzing the distribution of MIDI features given each token and the normalized mutual information (NMI) between tokens and features. We find that the tokens from the first codebook encode acoustic energy as well as related MIDI features such as velocity, but not more "musical" features such as pitch or polyphony, while later codebook tokens do not appear to encode any meaningful MIDI features.

1 INTRODUCTION AND RELATED WORK

Discrete audio tokenizers convert continuous audio waveforms into sequences of discrete tokens, a form of representation learning. Speech tokenizers are a particularly successful example of discrete audio tokenizers, especially as a representation model, with models such as Mimi using distillation to learn tokens that encode phonetic information (Défossez et al., 2024). Discrete audio tokenizers for music are also useful as they can be used for tasks like music generation such as Dieleman et al. (2018) tokenizing piano audio through an autoregressive discrete autoencoder before training a generative model on the learned tokens, which Dhariwal et al. (2020) similarly does except using a VQ-VAE for the tokenization and training on general music.

Despite the recent success of such models, little is known about what features the learned tokens encode. Prior work to study this includes Castellon et al. (2021), which showed that using transformer layer activations from the generative model in Jukebox helps to improve classifier performance on tasks such as genre classification and key detection. MERT (Li et al., 2024) used RVQ-VAE tokens as an acoustic teacher and combined this with a Constant-Q transform (CQT) based musical teacher for similar music understanding tasks as they suggest that RVQ-VAE tokens may emphasize acoustic features over musical features such as pitch or timbre. However, these studies evaluate what features the tokens encode primarily through downstream classifier performance, not by directly analyzing the tokens.

In this work, we distinguish between low-level acoustic properties such as volume and higher level musical features such as pitch and polyphony. We aim to evaluate what features the tokens from music audio tokenizers encode, and specifically whether the tokens encode musically meaningful information like tone, through direct analysis on the tokens.

2 METHODOLOGY

We train an RVQ-VAE on the MAESTRO dataset of piano audio, then using the aligned MIDI annotations, which provide ground truth labels of what the MIDI features are at each point in time,

we can find the distribution of what MIDI features occur when each token is used and calculate the NMI between tokens and features across codebooks.

2.1 DATASET

The dataset is a subset of the MAESTRO dataset (v3.0.0) from Google’s Magenta, which consists of audio recordings of classical piano performances done on electronic pianos stored in wav files, along with the MIDI labels for each recording from the electric piano which stores information such as what note was played at each time, the velocity it was played at, and others.

We split the wav recordings into 4 second clips, convert the audio into single channel by averaging the stereo channels together, resample it to a sample rate of 16000 Hz, then normalize the audio. Each 4 second clip becomes a raw waveform represented by a one dimensional vector with 64000 entries. We used the train/validation/test split recommended by the dataset.

2.2 MODEL

The model is trained on these raw waveforms with an architecture inspired by EnCodec (Défossez et al., 2022) and Dieleman et al. (2018). We use an RVQ-VAE with 8 codebooks of 512 entries each.

The encoder portion consists of a 1D convolution to project from 1 to 32 channels, then a sequence of 3 convolutional blocks, each doubling the number of channels and halving the temporal length, followed by a 1D convolution to project the number of channels down to 128, producing a 128×8000 tensor for each 4 second sound clip.

Each 128×1 vector in the tensor represents a $1/8000$ clip of the input clip (0.5 ms) and we pass these as input to 8 vector quantizers in sequence. Each quantizer takes the vector and maps it to the nearest entry for the corresponding codebook, then we pass the residual to the next quantizer. This results in each 0.5 ms of the input sequence represented by a vector of 8 token indices, one from each codebook.

The decoder mirrors the encoder, with a 1D convolution to expand from 128 to 256 channels followed by transposed convolutional blocks, each halving the number of channels and doubling the temporal length, a final 1D convolution to project back to 1 channel, and a Tanh activation to normalize the output to $[-1, 1]$.

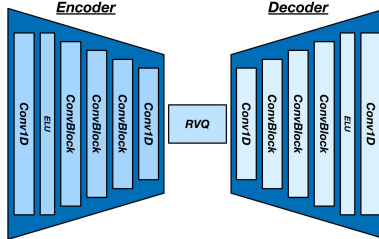


Figure 1: RVQ-VAE architecture. The encoder maps raw waveforms into a latent representation that is successively quantized by 8 residual codebooks. Each codebook quantizes the residual left by the previous codebooks, and the decoder reconstructs the waveform from the resulting discrete representation.

Our loss is inspired by EnCodec, which uses the L^1 loss between the original and reconstructed waveforms, mel-spectrogram loss to capture frequency, a discriminator loss, and a codebook commitment loss to prevent codebook collapse. However, we omit the discriminator loss given time and resource constraints. We also replace the L^1 loss with SI-SNR loss, which we found to empirically improve reconstruction quality. Specifically, the loss we use is $\mathcal{L}(\hat{x}, x) = \mathcal{L}_{\text{SI-SNR}}(\hat{x}, x) + \mathcal{L}_{\text{mel}}(\hat{x}, x) + \mathcal{L}_{\text{commit}}$, where

$$\mathcal{L}_{\text{SI-SNR}} = -\text{SI-SNR}(\hat{x}, x),$$

$$\mathcal{L}_{\text{mel}} = \sum_{i \in e} \left(\mathcal{L}_{\text{sc}}^{(i)} + \|S_i(\hat{x}) - S_i(x)\|_1 + \|S_i(\hat{x}) - S_i(x)\|_2 \right),$$

$$\mathcal{L}_{\text{commit}} = \sum_{i=1}^N \|z_i - q_i(z_i)\|_2^2,$$

where e is the set of window sizes, $S_i(x)$ represents the mel-spectrogram of x , and the spectral loss is $\mathcal{L}_{\text{sc}}^{(i)} = \frac{\|S_i(\hat{x}) - S_i(x)\|_F}{\|S_i(x)\|_F}$ where $\|\cdot\|_F$ is the Frobenius norm. For the commitment loss, z_i is the residual for the corresponding quantizer and $q_i(z_i)$ is the nearest entry in the corresponding codebook.

2.3 ANALYSIS

We use the test set to conduct analysis on token usage statistics, the distribution of features given a codebook token, and the NMI between codebook tokens and features, as well as to evaluate the reconstruction quality. We input every clip into the model and for each clip, we sampled the tokens used every 0.1 seconds along with the MIDI labels for that time from the aligned MIDI annotations. We sampled RMS volume at each sample time from the wav files as MIDI does not encode this information.

We calculated the perplexity of the token usage distribution for each codebook through $\text{PP}(X) = \exp(H(X))$ where $H(X) = -\sum_{i=1}^N p_i \ln p_i$, X is the token usage distribution for a given codebook, $H(X)$ is the entropy of the distribution, and p_i is the probability of the i th token appearing.

Using the distribution of token usage, distribution of each feature, and what the tokens and feature values were at each point in time, we calculated the NMI between tokens and features as well as the distribution of feature values given a token. For continuous valued features such as volume, we split the values into buckets for the NMI and distribution heatmap.

3 RESULTS

All results were evaluated on a held out test set not seen during training, tuning, and model selection.

3.1 RECONSTRUCTION QUALITY

The model evaluated on the test set achieved a SI-SNR of +33.00 dB and mel-spectrogram loss of 0.256. Although quantitative benchmarks for audio can be inaccurate compared to human perceived audio quality, SI-SNR is a well-established quantitative benchmark known to correlate with perceived audio quality and qualitative benchmarks were not feasible for this project due to time and resource constraints. As a baseline, the audio codec Opus at the same bitrate (144 kbps) achieves a SI-SNR loss of +36.47 dB on the same test set, indicating that our model has a reasonable audio reconstruction quality. Qualitatively, while the original audio sounds better than the reconstructed audio from our model, the difference is usually small.

3.2 TOKEN USAGE STATISTICS

Unique token count and token usage perplexity both increase for later codebooks, with the first codebook having a significantly lower number of unique tokens and perplexity than the rest. This suggests that the first codebook is more specialized than later codebooks, consistent with later findings. An exception was the third codebook which had a lower perplexity of 215.3 compared to the other later codebooks.¹

¹We calculated token usage by taking every 10th clip in the test set. The token rank sorted by usage for later plots was determined by this value, which is why some tokens that were never used according to the token usage statistics appear to be used. As the tokens affected were never used in a large random subset of the test set, this does not affect overall findings.

Table 1: Codebook utilization statistics across all 8 RVQ codebooks

CODEBOOK	UNIQUE TOKENS	PERPLEXITY
0	323/512	136.3
1	503/512	278.2
2	511/512	215.3
3	512/512	260.1
4	512/512	266.7
5	512/512	270.3
6	512/512	277.5
7	512/512	286.5

3.3 LATENT SPACE VISUALIZATION

We apply t-SNE on the codebook tokens, both on all codebook tokens together at once and the first and last codebooks individually, with the other intermediate codebooks omitted as they resemble the last codebook.

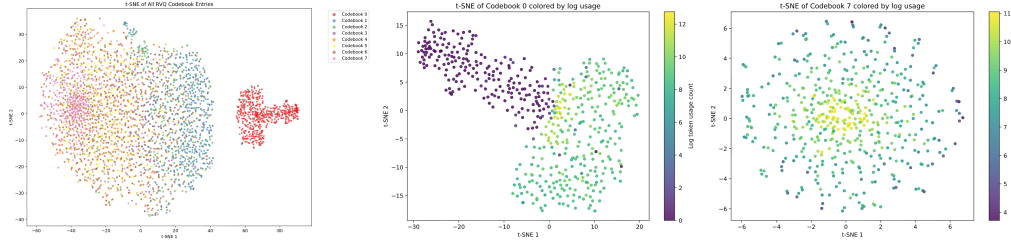


Figure 2: t-SNE of all codebook tokens colored by codebook index (left), t-SNE of first codebook colored by log usage (center), and t-SNE of last codebook colored by log usage (right).

We again observe that the first codebook is distinct, with its tokens appearing in their own cluster separate from the other codebooks in the t-SNE visualization of all the codebook tokens. From the visualization of the individual codebooks, the tokens in the first codebook are also more organized in the latent space, with clear clusters separating the most and least used tokens. While the last codebook also has the more used tokens clustered towards the center in the t-SNE visualization, it does not have the same clear separation as the first codebook.

3.4 FEATURE DISTRIBUTION GIVEN TOKEN

We use the distribution of different features given token appearances for different codebooks to plot heatmaps with the token index sorted by test set usage.

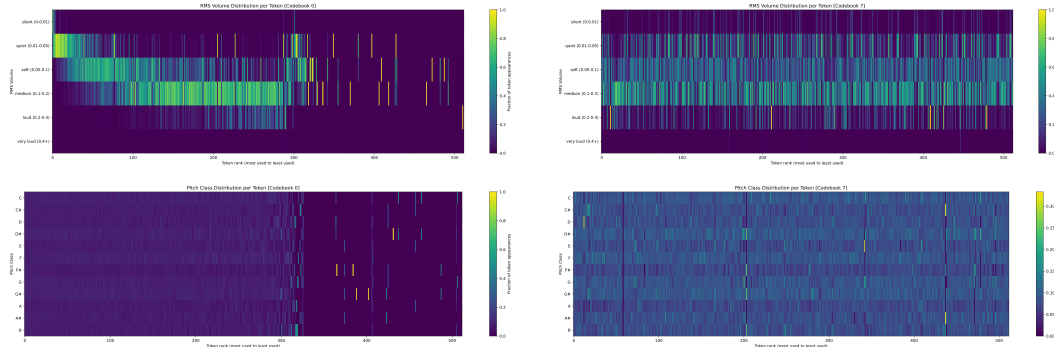


Figure 3: RMS volume distribution per token for the first codebook (top left) and last codebook (top right). Pitch class distribution per token for the first codebook (bottom left) and last codebook (bottom right). Token rank is sorted by usage frequency.

The tokens in the first codebook show a narrow volume distribution per token while the volume distribution per token for the last codebook is uniform. A similar pattern appears for the velocity and temporal range per token, which are closely related to volume. For pitch class and other features unrelated to volume, the distribution is uniform per token across all codebooks. This suggests that the first codebook encodes acoustic energy and related features, but not any features unrelated to acoustic energy, while later codebooks do not appear to strongly encode any MIDI features.

3.5 NORMALIZED MUTUAL INFORMATION

Normalized mutual information between tokens across codebooks and features shows a similar pattern where volume-related features are encoded in the first and earlier codebooks, while limited features are encoded in later codebooks.

Table 2: Normalized mutual information between codebook tokens and MIDI features across all 8 codebooks.

Codebook	RMS	Velocity	Polyphony	Time	Pitch Class	Octave	Pitch	Pedal
0	0.180	0.058	0.001	0.006	0.006	0.013	0.017	0.001
1	0.148	0.053	0.002	0.006	0.006	0.009	0.014	0.001
2	0.141	0.056	0.002	0.005	0.005	0.008	0.013	0.001
3	0.124	0.052	0.002	0.004	0.005	0.007	0.012	0.001
4	0.117	0.051	0.001	0.004	0.004	0.006	0.012	0.001
5	0.104	0.048	0.001	0.003	0.004	0.006	0.011	0.001
6	0.097	0.047	0.001	0.003	0.003	0.005	0.011	0.001
7	0.087	0.045	0.002	0.002	0.003	0.005	0.011	0.001

4 CONCLUSION

The increasing perplexity across codebooks along with the t-SNE visualizations showing a separate cluster for the first codebook, which also had a more structured geometry when viewed individually, suggests that tokens from the first codebook capture meaningful specialized information. From the distribution of volume and related features per token and the NMI between tokens and volume, we conclude that tokens from the first codebook encode acoustic energy while later codebooks may encode information not well captured by any MIDI features. These findings are consistent with MERT’s assumption that RVQ-VAE based music tokenizers mainly encode acoustic information rather than more musical information that would be captured by MIDI features such as pitch or polyphony.

FUTURE WORK

Future work could evaluate what information the later codebooks encode. For example, finding the feature distribution and NMI analysis while controlling for volume could reveal new correlations; as the NMI with RMS decreased by over 50% between the first and last codebook while the NMI for velocity remained relatively constant, it is possible that the later codebooks encode tone in some way that is related to velocity.

To push the tokenizer to learn more musical tokens, the loss could be changed to include a CQT-based loss which would emphasize pitch for the reconstruction. Similar analysis could also be done by training GAN based models such as EnCodec on the same dataset to see if the discriminator may help with learning more musically meaningful token representations.

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